School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Milestone - 3 Scribe

Eigenvalue-Based Cooperative UAV Target Detection

Q-1: System Objective and Deterministic Baseline

Answer: Project Objective

The main objective of this project is to analyze how multiple UAVs can cooperate to detect targets in a surveillance region under uncertain sensing conditions. In real environments, UAV sensors receive signals that may contain both the target signal and environmental noise. Because of this uncertainty, the system must determine whether a target is present or not using probabilistic modeling.

The detection problem can be represented as a binary hypothesis testing problem:

H_0 : Noise only
H_1 : Target signal + Noise

The system aims to achieve:

- High Probability of Detection (P_d)
- Low Probability of False Alarm (P_{fa})

Deterministic Baseline Strategy

Instead of using optimization algorithms, UAVs are placed according to a fixed rule. This deterministic rule provides a simple reference model to understand system behavior.

The surveillance region is defined as:

$$\text{Area size} = 5000\,m \times 5000\,m$$

However, UAVs are allowed to be deployed only in the deployable region:

$$0 \leq X \leq 1800$$

Within this region, UAVs are positioned using a deterministic rule.

All UAVs are placed at a fixed X coordinate:

$$X = \frac{DEPLOYABLE_X_MAX}{2} = 900$$

UAVs are then evenly distributed along the Y-axis.

The deterministic placement is implemented in the simulation code (shown below).

```
def baseline_placement(M):

    uavs = np.zeros((M, 2))
    fixed_x = DEPLOYABLE_X_MAX / 2
    y_spacing = AREA_SIZE / (M + 1)

    for i in range(M):
        uavs[i, 0] = fixed_x
        uavs[i, 1] = y_spacing * (i + 1)

    return uavs
```

This produces evenly spaced UAV locations across the sensing region.

Three scenarios are evaluated:

- $M = 3$ UAVs
- $M = 4$ UAVs
- $M = 5$ UAVs

This deterministic strategy is reasonable because evenly spaced deployment provides balanced coverage and is commonly used as a baseline configuration in sensor network studies.

Q-2: Random Variables in the Simulation

Answer: Although UAV placement is deterministic in this milestone, several components of the sensing system still contain uncertainty.

1. Environmental Noise

Environmental noise affects the signal received by UAV sensors.

$$N \sim \mathcal{N}(0, \sigma^2)$$

This Gaussian noise represents interference from the wireless environment.
Noise is added to the received signal and introduces uncertainty in the sensing process, affecting the detection statistics used by the UAV.

2. Target Signal

The signal transmitted by the target is also uncertain due to channel variations.

$$S \sim \mathcal{N}(0, \sigma_s^2)$$

When a target is present, the received signal becomes

$$Y = S + N$$

Otherwise, only noise is observed.

3. Target Position

Targets may appear at different locations in the sensing region. In the simulation, target positions may either be fixed or randomly generated.

$$X_t \sim \text{Uniform}(1800, 5000)$$

$$Y_t \sim \text{Uniform}(0, 5000)$$

These distributions represent random target placement outside the UAV deployable region.

Q-3: Probabilistic Modeling and Fixed Decision Rule

Answer: Uncertainty in our system is primarily caused by the wireless sensing environment especially noise and variation in signals. Although the location of the UAVs and the position of the antennas in the simulation would be determined by a deterministic rule, the process of sensing has probabilistic parts.

Probabilistic Assumptions

Additive white Gaussian noise (AWGN) affects the received signal of every UAV, which brings about randomness in the measurements. Due to such noise, the statistic of detection is a random variable.

The probability of detection depends on the signal-to-noise ratio (SNR) between the UAV and the target. For the eigenvalue-based detector used in this project, the probability of detection can be expressed as:

$$P_d = Q\left(Q^{-1}(P_{fa}) - \frac{SNR\sqrt{K/2}}{1+SNR}\right)$$

where P_{fa} represents the probability of false alarm and K represents the number of collected samples.

Deterministic Baseline Under Uncertain Inputs

Although the sensing model contains uncertainty, the baseline simulation uses a deterministic rule for UAV behavior. Each UAV calculates the distance to all targets and selects the nearest target for antenna orientation.

```
target_dists = [np.linalg.norm(u - t) for t in targets]
nearest_t = targets[np.argmin(target_dists)]
```

Since the nearest target is always selected using a fixed rule, the decision mechanism remains deterministic even when the sensing environment contains uncertainty.

Sum Detection Probability

To evaluate the overall sensing performance of the UAV network, we compute the sum detection probability across all targets:

$$P_{sum} = \sum_{n=1}^{N} P_n$$

where P_n represents the detection probability for target n. In a cooperative UAV sensing system, the detection probability for each target depends on all UAVs observing that target and can be expressed as:

$$P_n = 1 - \prod_{m=1}^{M} (1 - P_{m,n})$$

where $P_{m,n}$ represents the detection probability of UAV m for target n.

Q-4: Simulation Pipeline

Answer: Input Generation

The simulation begins by defining the main system parameters:

- AREA_SIZE = 5000
- DEPLOYABLE_X_MAX = 1800
- MIN_UAV_SEPARATION = 500
- MIN_TARGET_SEPARATION = 400

Targets are initially defined at fixed locations within the sensing region:

- $(2600, 3500)$
- $(4000, 3800)$
- $(3000, 2000)$

These targets are intentionally spread across the sensing area in order to test detection coverage under different spatial configurations.

UAV Deployment

UAV positions are generated using the deterministic baseline placement function described earlier. The simulation evaluates three deployment scenarios with different numbers of UAVs.

```
print("Running Baseline Placement for M=3...")
best_uavs_3 = baseline_placement(3)
print("Running Baseline Placement for M=4...")
best_uavs_4 = baseline_placement(4)
print("Running Baseline Placement for M=5...")
best_uavs_5 = baseline_placement(5)
```

These calls generate UAV deployments for:

- $M = 3$ UAVs
- $M = 4$ UAVs
- $M = 5$ UAVs

Target Distance Computation

For each UAV, the distance to every target is computed using the Euclidean distance formula:

$$d = \sqrt{(x_u - x_t)^2 + (y_u - y_t)^2}$$

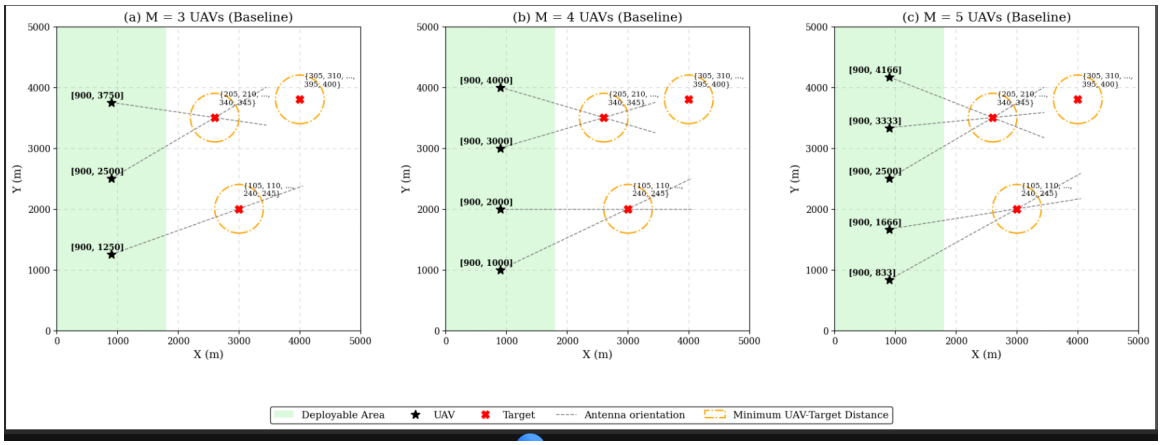
The nearest target is then identified and used to determine the sensing direction of the UAV.

Antenna Orientation

Each UAV antenna is oriented toward its nearest target. In the simulation visualization, this orientation is represented using dashed lines extending from the UAV location toward the target.

Visualization

The simulation produces a deployment diagram that illustrates the spatial configuration of the sensing system.



Baseline UAV deployment for $M = 3$, $M = 4$, and $M = 5$.

Output Metrics

The simulation produces several metrics used to evaluate sensing performance:

- UAV–target distances
- Detection probability for each target
- Sum detection probability across all targets
- Deployment visualization diagrams

These outputs allow us to analyze how UAV placement influences detection coverage and overall sensing performance.

Simulation code

```python
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches

AREA_SIZE = 5000
DEPLOYABLE_X_MAX = 1800
MIN_UAV_SEPARATION = 500
MIN_TARGET_SEPARATION = 400

targets = np.array([(2600, 3500), (4000, 3800), (3000, 2000)])

def baseline_placement(M):

    uavs = np.zeros((M, 2))
    fixed_x = DEPLOYABLE_X_MAX / 2
    y_spacing = AREA_SIZE / (M + 1)

    for i in range(M):
        uavs[i, 0] = fixed_x
        uavs[i, 1] = y_spacing * (i + 1)

    return uavs


print("Running Baseline Placement for M=3...")
best_uavs_3 = baseline_placement(3)
print("Running Baseline Placement for M=4...")
best_uavs_4 = baseline_placement(4)
print("Running Baseline Placement for M=5...")
best_uavs_5 = baseline_placement(5)

scenarios = [
    {"label": "(a)", "title": "M = 3 UAVs (Baseline)", "uavs": best_uavs_3},
    {"label": "(b)", "title": "M = 4 UAVs (Baseline)", "uavs": best_uavs_4},
    {"label": "(c)", "title": "M = 5 UAVs (Baseline)", "uavs": best_uavs_5},
```

```python
]

plt.rcParams.update({
    'font.size': 10,
    'axes.labelsize': 12,
    'axes.titlesize': 14,
    'legend.fontsize': 11,
    'font.family': 'serif'
})

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

for ax, scenario in zip(axes, scenarios):
    ax.set_xlim(0, AREA_SIZE)
    ax.set_ylim(0, AREA_SIZE)
    ax.set_xlabel("X (m)")
    ax.set_ylabel("Y (m)")
    ax.set_title(f"{scenario['label']} {scenario['title']}")
    ax.set_aspect('equal')
    ax.grid(True, linestyle=(0, (5, 5)), alpha=0.5)

    deploy_patch = patches.Rectangle((0, 0), DEPLOYABLE_X_MAX, AREA_SIZE,
                                     linewidth=0, facecolor='lightgreen', alpha=0.3, label='Deployable Area')
    ax.add_patch(deploy_patch)

    for idx, t in enumerate(targets):
        tx, ty = t[0], t[1]
        ax.plot(tx, ty, 'rX', markersize=8, label='Target' if idx == 0 else "")

        labels_list = ["{205, 210, ...,\n340, 345}", "{305, 310, ...,\n395, 400}", "{105, 110, ...,\n240, 245}"]
        ax.annotate(labels_list[idx % len(labels_list)], (tx, ty), textcoords="offset points", xytext=(5, 15), fontsize=8, color='black')

        min_dist_circle = patches.Circle((tx, ty), MIN_TARGET_SEPARATION, linewidth=1.5, edgecolor='orange',
                                         facecolor='none', linestyle='-.', label='Minimum UAV-Target Distance' if idx == 0 else "")
```

```python
        ax.add_patch(min_dist_circle)

    for idx, u in enumerate(scenario['uavs']):
        ux, uy = u[0], u[1]

        target_dists = [np.linalg.norm(u - t) for t in targets]
        nearest_t = targets[np.argmin(target_dists)]
        dx = nearest_t[0] - ux
        dy = nearest_t[1] - uy
        norm = np.hypot(dx, dy)
        if norm == 0: norm = 1

        ax.plot([ux, ux + dx * 1.5],
                [uy, uy + dy * 1.5],
                linestyle='--', color='grey', linewidth=1,
                label='Antenna orientation' if idx == 0 else "", zorder=1)

        ax.plot(ux, uy, 'k*', markersize=10, label='UAV' if idx == 0 else "", zorder=3)
        ax.annotate(f"[{int(ux)}, {int(uy)}]", (ux, uy), textcoords="offset points", xytext=(-45, 5), fontsize=9, fontweight='bold')

handles, labels = axes[0].get_legend_handles_labels()

order_of_items = ['Deployable Area', 'UAV', 'Target', 'Antenna orientation', 'Minimum UAV-Target Distance']
by_label = dict(zip(labels, handles))
ordered_handles = [by_label[k] for k in order_of_items if k in by_label]
ordered_labels = [k for k in order_of_items if k in by_label]

fig.legend(ordered_handles, ordered_labels, loc='lower center', ncol=5, bbox_to_anchor=(0.5, -0.05), frameon=True, edgecolor='black')

plt.tight_layout()
fig.subplots_adjust(bottom=0.2)
plt.savefig("baseline_uav_deployment_simulation.png", dpi=300, bbox_inches='tight')
print("Saved baseline_uav_deployment_simulation.png")
plt.close()
```

Q-5: Simulation Outputs and Interpretation

Answer:

Performance Metrics

The Sum Detection Probability is the main performance metric used in the simulation, which represents the overall detection capability of the UAV network across all targets.

This metric measures how effectively the UAV network can detect multiple targets simultaneously. The greater the values of the sum detection probability, the higher the sensing performance.

In addition to this metric, the simulation also analyzes:

- Distances between UAVs and targets
- UAV spatial deployment patterns
- Antenna orientation toward the nearest targets

Statistical Summaries and Visualization

Simulation produces deployment visualization diagrams of the UAVs positions in the sensing region. These diagrams include:

- UAV positions
- Target locations
- Deployable UAV region
- Antenna orientation lines
- Minimum UAV–target distance constraints

Observed Patterns Under the Deterministic Strategy

There are a number of trends that may be noticed based on the deterministic simulation of a baseline:

- The more the number of UAVs, the higher the spatial coverage of the sensing area.
- UAVs will always make the antennae face the closest objects and this enables the enhancement of sensing.
- Targets that are closer to UAVs generally have higher detection probability due to stronger signal reception.

- The UAVs should be evenly distributed to achieve balanced coverage but not necessarily to give the optimal sensing configuration.

Q-6: Technical Gaps and Preparation for Next Stage

Answer: Limitations of the Deterministic Baseline

The current simulation uses a fixed rule for UAV placement, where UAVs are positioned at a constant X-coordinate and evenly spaced along the Y-axis.
Although this gives a balanced coverage, this does not assure optimal placement to maximize detection probability.

Role of Randomness in Future Improvements

Randomized algorithms have the capability to search a variety of UAV configurations rather than using one deterministic rule.
This may be used to locate improved UAV positions and enhance overall detection.

Preparation for Milestone 4

In Milestone 4, we will study and implement a randomized algorithm to optimize UAV deployment.
This approach will help us find better UAV positions for target detection and compare the results with the current deterministic baseline.